

MITRE ATLAS Overview

ATLAS stands for Adversarial Threat Landscape for Artificial-Intelligence Systems.

This document is designed as clean, selectable text for a RAG knowledge base.

What MITRE ATLAS Is

MITRE ATLAS is a publicly accessible knowledge base of adversary tactics and techniques against AI-enabled systems. It organizes real-world observations and realistic demonstrations into a structured framework that security teams can use for threat modeling, investigations, red teaming, and defensive planning.

Why It Matters

AI systems introduce risks involving training data, model behavior, model files, inference interfaces, third-party components, and supporting pipelines. ATLAS gives defenders a common vocabulary for describing those risks.

Common AI Security Risks

Risk	Description
Data poisoning	Changing or adding training data to influence model behavior, reduce accuracy, or create hidden behavior.
Model evasion	Crafting inputs that cause incorrect classifications or unsafe model behavior.
Model extraction	Using repeated queries or unauthorized access to approximate or steal a model.
Model inversion	Attempting to infer sensitive information associated with model outputs or training data.

Prompt injection	Supplying instructions intended to override expected behavior or expose restricted information.
Supply-chain compromise	Compromising models, datasets, packages, plugins, repositories, or development infrastructure.
Abuse of AI capabilities	Using AI to increase the speed, scale, or effectiveness of malicious activity.

How ATLAS Is Organized

A tactic represents an adversary objective. A technique describes how the adversary attempts to achieve that objective. Case studies document observed or demonstrated activity, while mitigations describe defensive measures that can reduce risk.

SOC Uses

- Classify suspicious activity involving AI applications, models, APIs, repositories, or training pipelines.
- Map logs and alerts to AI-focused adversary techniques.
- Identify mitigations for prompt injection, model theft, poisoning, evasion, and supply-chain risks.
- Improve incident reports with consistent AI-security terminology.

Investigation Workflow

1. Identify the affected model, application, API, dataset, or pipeline.
2. Collect prompts, responses, access logs, model changes, package changes, identity events, and network evidence.
3. Map the evidence to an ATLAS tactic and technique.
4. Assess impact on confidentiality, integrity, availability, privacy, safety, and reliability.
5. Contain the issue by restricting access, disabling integrations, isolating assets, or reverting altered data and models.
6. Apply mitigations and document the mapped ATLAS behavior.

Relationship to MITRE ATT&CK

MITRE ATT&CK describes adversary behavior across conventional enterprise, mobile, and industrial-control environments. ATLAS complements ATT&CK by focusing on AI-enabled systems. One incident may use both frameworks.

Suggested RAG Questions

- What is MITRE ATLAS?
- How does ATLAS differ from MITRE ATT&CK?
- What is data poisoning?
- How should a SOC investigate prompt injection?
- How can an organization reduce model-extraction risk?

References

MITRE ATLAS: <https://atlas.mitre.org/>

MITRE ATT&CK: <https://attack.mitre.org/>